Point Estimation:

Goal: To find the MVUE for a parameter θ.

1) Find what statistic(s) are minimally sufficient through factorization.

2) Formulate an unbiased estimator as a function of the minimally sufficient statistics.

$\Rightarrow$ The result will be the MVUE.

e.g. Consider $Y_1, \dots, Y_n \sim Exp(\beta)$. $\mu_Y = \beta$

$$L(\beta) = \prod_{i=1}^{n} \frac{1}{\beta} e^{-y_i/\beta} = \underbrace{\frac{1}{\beta^n} e^{-\frac{1}{\beta}\sum y_i}}_{g(u,\beta)} \times \underbrace{1}_{h(y_1,\dots,y_n)}$$

1) $U = \sum^{n} Y_i$ is minimally sufficient for β.

2) $\frac{\sum_{i=1}^{n} Y_i}{n} = \bar{Y}$ is the MVUE.

Note: $e^{-\frac{1}{\beta}\sum y_i} = e^{-\frac{1}{\beta} n \bar{y}}$

$\bar{Y}$ is minimally sufficient.

Stat 266 – Unit 8 – Video Set 6 – Class Examples

Summary:

Rao-Blackwell Theorem and MVUE

Find U by factorization.

Theorem 9.5 : Rao - Blackwell Theorem

Let $\hat{\theta}$ be any unbiased estimator for θ where $V(\hat{\theta}) < \infty$.

If U is a sufficient statistic for θ, define $\hat{\theta}^* = E[\hat{\theta} \mid U]$.

Then for all θ,

$E[\hat{\theta}^*] = \theta$ and $V[\hat{\theta}^*] \le V[\hat{\theta}]$.

1) Is $\hat{\theta}^*$ a statistic?

- If U is sufficient then the conditional likelihood $(L(\hat{\theta} \mid U))$ does not depend on θ.
- Then $E[\hat{\theta} \mid U] = \underbrace{\hat{\theta}^*}$ cannot depend on θ.

  a function of $Y_1, \ldots, Y_n$ and other known constants.

2) $E[\hat{\Theta}^*] = E[E[\hat{\Theta} \mid U]] = E[\hat{\Theta}] = \Theta$

3) $V[\hat{\Theta}] = E[V[\hat{\Theta} \mid U]] + V(\underbrace{E[\hat{\Theta} \mid U]}_{\hat{\Theta}^*})$

$\Rightarrow V(\hat{\Theta}^*) = V(\hat{\Theta}) - \underbrace{E[\underbrace{V(\hat{\Theta} \mid U)}_{\geq 0}]}_{\geq 0}$

$\Rightarrow V(\hat{\Theta}^*) \leq V(\hat{\Theta})$

Example A:

Suppose $X_1, X_2, \ldots, X_n$ is a random sample if independent and identically distributed $Bernoulli(p)$ random variables. Consider the unbiased estimator $\hat{p} = X_1$ for p, and the minimally suffient statistic $U = \sum_{i=1}^{n} X_i$. Use the Rao-Blackwell theorem to find the $MVUE$ for p.

the first observation

Parameter: $\theta = p$

$P(X = x) = p^x (1-p)^{1-x}, \quad x = 0, 1$

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = \underbrace{p^{\sum x_i}(1-p)^{n-\sum x_i}}_{g(u,p)} \times \underbrace{1}_{h(y_1,\ldots,y_n)}$$

So $U = \sum_{i=1}^{n} X_i$ is minimally sufficient for p.

So $\hat{\theta} = \frac{U}{n} = \frac{\sum^{n} X_i}{n} = \bar{X} = \hat{p}$ is the MVUE for p.

Rao-Blackwell: Start with $\hat{p} = X_1$ $\quad (E[\hat{p}] = E[X_1] = p)$

$$\cdot\ \hat{p}^* = E[\hat{p} \mid U] = E[X_1 \mid \sum_{i=1}^{n} X_i = u]$$

$$= \sum_{x=0}^{1} x\, P(X_1 = x \mid \sum_{i=1}^{n} X_i = u) = P(X_1 = 1 \mid \sum_{i=1}^{n} X_i = u)$$

$$= \frac{P(X_1 = 1, \sum_{i=1}^{n} X_i = u)}{P(\sum^{n} X_i = u)} = \frac{P(X_1 = 1)\, P(\sum_{i=1}^{n} X_i = u \mid X_1 = 1)}{P(\sum_{i=1}^{n} X_i = u)}$$

$$= \frac{P(X_1 = 1)\, P(\sum_{i=2}^{n} X_i = u - 1)}{P(\underbrace{\sum_{i=1}^{n} X_i}_{\sim\, Bin(n,p)} = u)} \quad \sim Bin(n-1, p)$$

$$= \frac{p \binom{n-1}{u-1} p^{u-1} (1-p)^{n-u}}{\binom{n}{u} p^{u} (1-p)^{n-u}} = \frac{\binom{n-1}{u-1}}{\binom{n}{u}} = \frac{\frac{(n-1)!}{(u-1)!\,(n-u)!}}{\frac{n!}{u!\,(n-u)!}}$$

$$= \frac{(n-1)!}{(u-1)!} \cdot \frac{u!}{n!} = \frac{u}{n} = \frac{\sum_{i=1}^{n} X_i}{n} = \bar{X}.$$

Example B:

Suppose $X_1, X_2, \ldots, X_n$ is a random sample if independent and identically distributed $Poisson(\lambda)$ random variables. Find the $MVUE$ for $\theta = P(X=0) = e^{-\lambda}$. $\lambda = \mu_Y$

Parameter: $e^{-\lambda}$

$\bar{Y}$ is the MVUE for λ.

Try $\hat{\theta} = e^{-\bar{Y}}$.

$$E[\hat{\theta}] = E[e^{-\bar{Y}}] = \cdots = e^{-\lambda(n - ne^{-1/n})} \neq e^{-\lambda}$$

So $e^{-\bar{Y}}$ is biased.

To use Rao-Blackwell, we need to start with some unbiased estimator.

* Consider $\hat{\theta} = Y = \begin{cases} 1, & \text{if } X_1 = 0 \\ 0, & \text{otherwise} \end{cases}$ the first observation.

$$E[\hat{\theta}] = 1P(X_1 = 0) + 0P(X_1 \neq 0) = P(X_1 = 0)$$
$$= e^{-\lambda}.$$

$\hat{\theta}$ is unbiased.

- $U = \sum_{i=1}^{n} X_i$ is minimally sufficient.

- $\hat{\Theta}^* = E[\hat{\Theta} \mid \sum_{i=1}^{n} X_i = u] = E[Y \mid \sum_{i=1}^{n} X_i = u]$

$$= 1\, P(Y = 1 \mid \sum_{i=1}^{n} X_i = u) + 0\, P(Y = 0 \mid \sum_{i=1}^{n} X_i = u)$$

$$= P(X_1 = 0 \mid \sum_{i=1}^{n} X_i = u) = \frac{P(X_1 = 0, \sum_{i=1}^{n} X_i = u)}{P(\sum_{i=1}^{n} X_i = u)}$$

$$= \frac{P(X_1 = 0)\, P(\sum_{i=1}^{n} X_i = u \mid X_1 = 0)}{P(\sum_{i=1}^{n} X_i = u)}$$

$$= \frac{P(X_1 = 0)\, P(\sum_{i=2}^{n} X_i = u)}{P(\sum_{i=1}^{n} X_i = u)} = \frac{e^{-\lambda} \frac{((n-1)\lambda)^u e^{-(n-1)\lambda}}{u!}}{\frac{(n\lambda)^u e^{-n\lambda}}{u!}}$$

$X_1 \sim \text{Poisson}(\lambda)$

$\sum_{i=2}^{n} X_i \sim \text{Poisson}((n-1)\lambda)$

$\sum_{i=1}^{n} X_i \sim \text{Poisson}(n\lambda)$

$$= \left(\frac{n-1}{n}\right)^u = \left(\frac{n-1}{n}\right)^{\sum_{i=1}^{n} X_i}.$$

$\Leftarrow$ The MVUE for $\Theta = e^{-\lambda}$.

<u>Check:</u> $E\left[\left(\frac{n-1}{n}\right)^{\sum_{i=1}^{n} X_i}\right] = E\left[e^{\sum_{i=1}^{n} X_i \cdot \ln\left(\frac{n-1}{n}\right)}\right]$

$$\overset{\text{iid}}{=} \left(E\left[e^{X \ln\left(\frac{n-1}{n}\right)}\right]\right)^n = \left[M_X\left(\ln\left(\frac{n-1}{n}\right)\right)\right]^n$$

$$= \left[e^{\lambda\left(e^{\ln\left(\frac{n-1}{n}\right)} - 1\right)}\right]^n = \left[e^{\lambda\left(\frac{n-1}{n} - 1\right)}\right]^n$$

$$= e^{n\lambda\left(\frac{n-1}{n} - 1\right)} = e^{-\lambda}. \quad \checkmark$$

Example C:

Suppose $X_1, X_2, \ldots, X_n$ is a random sample if independent and identically distributed $Poisson(\lambda)$ random variables. Find the $MVUE$ for $\theta = [P(X=0)]^2 = e^{-2\lambda}$.

- $U = \sum_{i=1}^{n} X_i$ is minimally sufficient.
- We need some unbiased estimator for $e^{-2\lambda}$.
- Consider $g(X)$ to be a function of a single observation. We need $E[g(X)] = e^{-2\lambda}$.

$$\Rightarrow E[g(X)] = \sum_{x=0}^{\infty} g(x) \cdot \frac{e^{-\lambda}\lambda^x}{x!} \Rightarrow e^{-2\lambda}$$

$$\Rightarrow \sum_{x=0}^{\infty} g(x) \frac{\lambda^x}{x!} = e^{-\lambda},$$

Note:

$$e^{-\lambda} = \sum_{x=0}^{\infty} \frac{(-1)^x \lambda^x}{x!}$$

The only unbiased estimator as a function of a single observation is (for $\theta = [P(X=0)]^2 = e^{-2\lambda}$):

$$\hat{\theta} = (-1)^{x_1} = \begin{cases} -1, & \text{if the first obs is odd } (x = 1, 3, 5, \ldots) \\ +1, & \text{" " " " " even } (x = 0, 2, 4 \ldots) \end{cases}$$

$$E\left[\hat{\Theta} \mid \sum_{i=1}^{n} X_i = u\right] = E\left[(-1)^{X_1} \mid \sum_{i=1}^{n} X_i = u\right]$$

$$= \sum_{x=0}^{\infty} (-1)^x P\left(X_1 = x \mid \sum_{i=1}^{n} X_i = u\right)$$

$$= \sum_{x=0}^{u} (-1)^x \frac{P(X_1 = x)\, P\left(\sum_{i=2}^{n} X_i = u - x\right)}{P\left(\sum_{i=1}^{n} X_i = u\right)}$$

$$= \cdots = \left(\frac{n-2}{n}\right)^u .$$

So $\left(\frac{n-2}{n}\right)^{\sum_{i=1}^{n} X_i}$ or $\left(\frac{n-2}{n}\right)^{n\bar{X}}$ is the MVUE

for $\Theta = [P(X=0)]^2 = e^{-2\lambda}$.

Example D:

Suppose $X_1, X_2, \ldots, X_n$ is a random sample if independent and identically distributed $Geometric(p)$ random variables. Find the $MVUE$ for p.

Is this unbiased?

$$E\left[\frac{n}{\sum X_i}\right] = ?$$

Recall: $\hat{p}_{MM} = \hat{p}_{MLE} = \frac{1}{\bar{X}} = \frac{n}{\sum_{i=1}^{n} X_i}$.

- $P(X = x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots$
- $L(p) = \prod_{i=1}^{n} p(1-p)^{x_i - 1} = \underbrace{p^n (1-p)^{\sum^n x_i - n}}_{g(u,p)} \times \underbrace{1}_{h(y_1, \ldots, y_n)}$
- $U = \sum_{i=1}^{n} X_i$ is minimally sufficient for p.
- We need some unbiased estimator:

Since $p = P(X = 1)$, Let $Y = \begin{cases} 1, & \text{if } X_1 = 1 \\ 0, & \text{otherwise} \end{cases}$.

$$E[Y] = 1P(X_1 = 1) + 0P(X_1 \geq 2)$$
$$= P(X_1 = 1) = p.$$

$$\hat{p}^* = E[\hat{p} \mid U = u] = E[Y \mid U = u] = P\left(X_1 = 1 \mid \sum_{i=1}^{n} X_i = u\right)$$

$$= \frac{P(X_1 = 1)\, P\left(\sum_{i=2}^{n} X_i = u - 1\right)}{P\left(\sum_{i=1}^{n} X_i = u\right)}$$

$\sum_{i=2}^{n} X_i \sim \text{NegBin}(r = n-1, p)$

$\sum_{i=1}^{n} X_i \sim \text{NegBin}(r = n, p)$

$$= \frac{\cancel{p} \binom{u-2}{n-2} \cancel{p^{n-1}} \cancel{(1-p)^{u-n}}}{\binom{u-1}{n-1} \cancel{p^{n}} \cancel{(1-p)^{u-n}}} = \frac{\dfrac{(u-2)!}{(n-2)!\,(u-n)!}}{\dfrac{(u-1)!}{(n-1)!\,(u-n)!}}$$

$$= \frac{(u-2)!}{(n-2)!} \cdot \frac{(n-1)!}{(u-1)!} = \frac{n-1}{u-1} = \frac{n-1}{\sum_{i=1}^{n} X_i - 1}.$$

So the MVUE for p is

$$\frac{n-1}{\sum_{i=1}^{n} X_i - 1} = \frac{1 - 1/n}{\bar{X} - 1/n},$$

$$\boxed{\hat{p}_{MM} = \hat{p}_{MLE} = \frac{1}{\bar{X}} \quad \Rightarrow \text{these are biased}}$$